

DATA VORTEX — PHASE 2

Social Engine Analytical Core

Phase 2 Insight Report

Analytical SQL, output evidence, logic explanation and insights

Database: DuckDB

Dataset: Social Engine cleaned dataset

Analysis period: May 2024 – April 2025

1. Schema Design

The cleaned Social Engine CSV files were converted into two analytical SQL tables: **users** and **posts**. DuckDB was used as the SQL engine.

Table	Rows	Purpose
users	1,500	User-level profile and account information
posts	8,712	Cleaned social-media posts and engagement metrics

Relationship: users.user_id → posts.user_id. One user can create multiple posts.

Key fields: post_id, user_id, platform, timestamp, likes, shares, comments and total_engagement.

2. SQL Queries

Challenge 1 — Trend Detection

```
-- Social Engine Phase 2 Analytical Core
```

```
-- Challenge 1: Trend Detection
```

```
SELECT strftime(timestamp, '%Y-%m') AS month, COUNT(*) AS post_count, ROUND(AVG(total_engagement), 2) AS avg_engagement FROM posts GROUP BY month
```

Challenge 2 — Platform Engagement Analysis

```
-- Challenge 2: Platform Engagement Analysis
```

```
SELECT platform, COUNT(*) AS post_count, ROUND(AVG(total_engagement), 2) AS avg_engagement, ROUND(MEDIAN(total_engagement), 2) AS median_engagement FROM posts GROUP BY platform
```

Challenge 3 — High-Engagement Detection

```
-- Challenge 3: High-Engagement Detection
```

```
WITH ranked AS (SELECT post_id, user_id, platform, total_engagement, ROW_NUMBER() OVER (PARTITION BY platform ORDER BY total_engagement DESC) AS rank) SELECT * FROM ranked WHERE rank < 10
```

Challenge 4 — Behavioural User Grouping

```
-- Challenge 4: Behavioural User Grouping
```

```
WITH user_stats AS (SELECT u.user_id, COUNT(p.post_id) AS post_count, ROUND(AVG(p.total_engagement), 2) AS avg_engagement FROM users u JOIN posts p ON u.user_id = p.user_id GROUP BY u.user_id) SELECT * FROM user_stats
```

Challenge 5 — Correlation Analysis

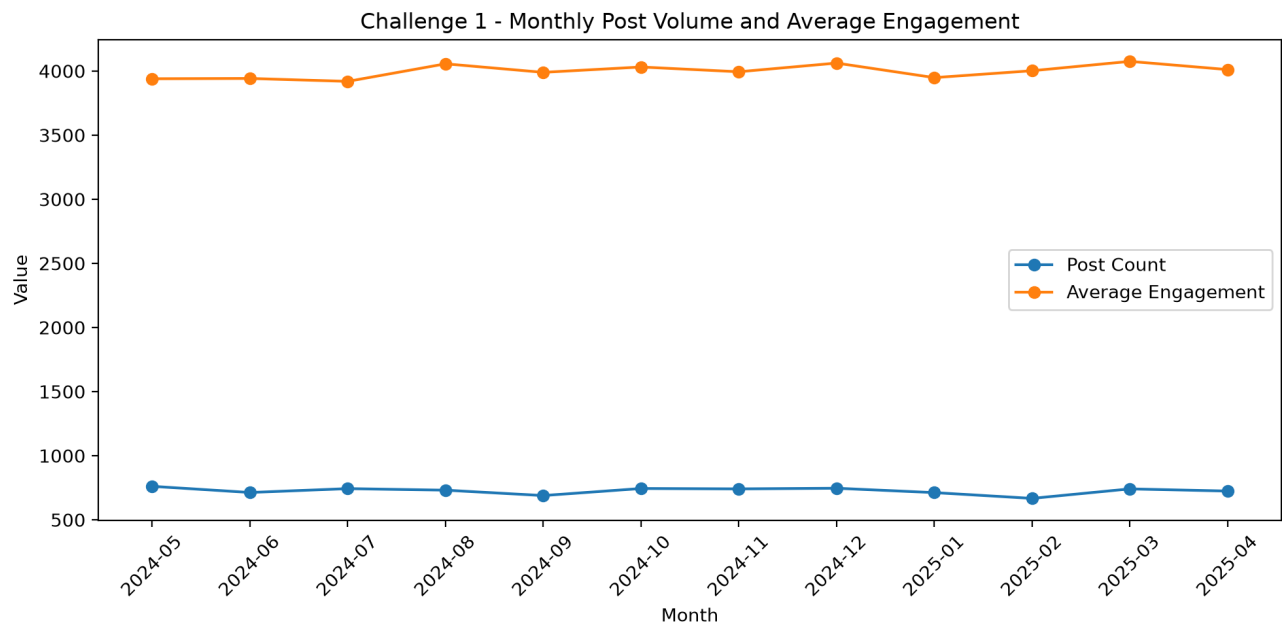
```
-- Challenge 5: Correlation Analysis
```

```
SELECT ROUND(corr(likes, shares), 4) AS likes_shares_corr, ROUND(corr(likes, comments), 4) AS likes_comments_corr, ROUND(corr(shares, comments), 4) AS shares_comments_corr FROM posts
```

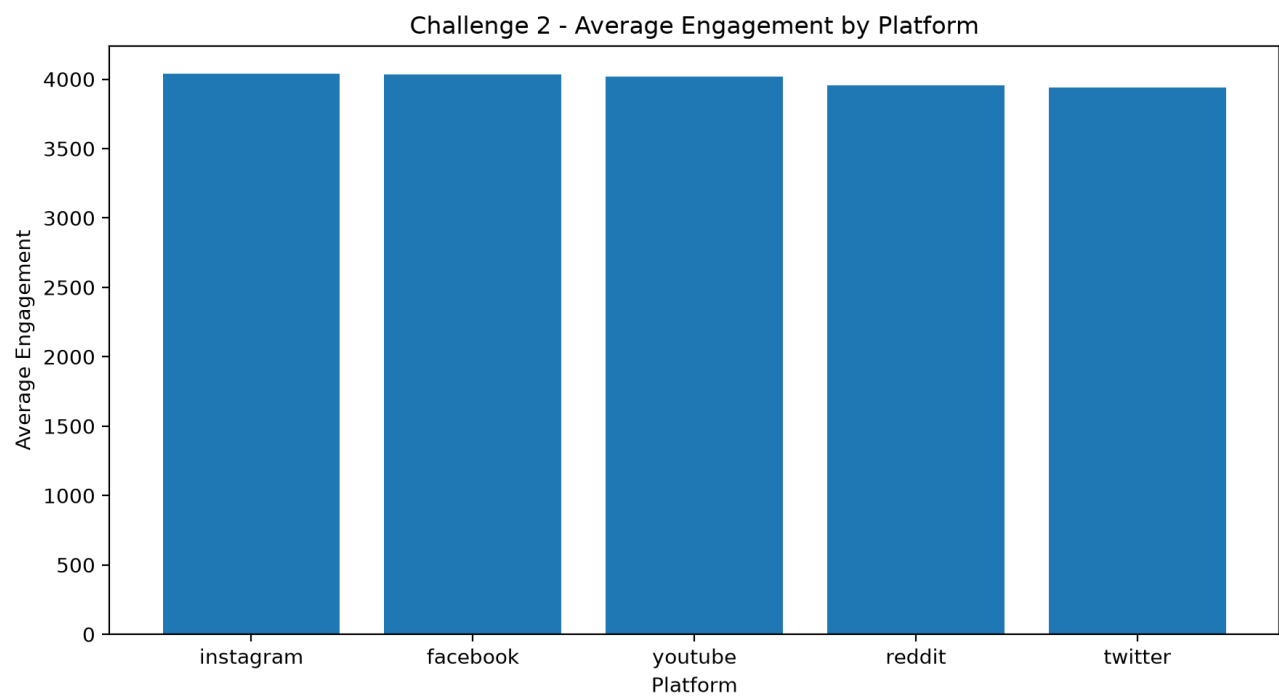
3. Output Screenshots

All outputs below were generated by executing the SQL queries directly against the DuckDB database. No analytical result was manually hardcoded.

Challenge 1 — Monthly Trend



Challenge 2 — Platform Engagement



Challenge 3 — Top 10 High-Engagement Posts

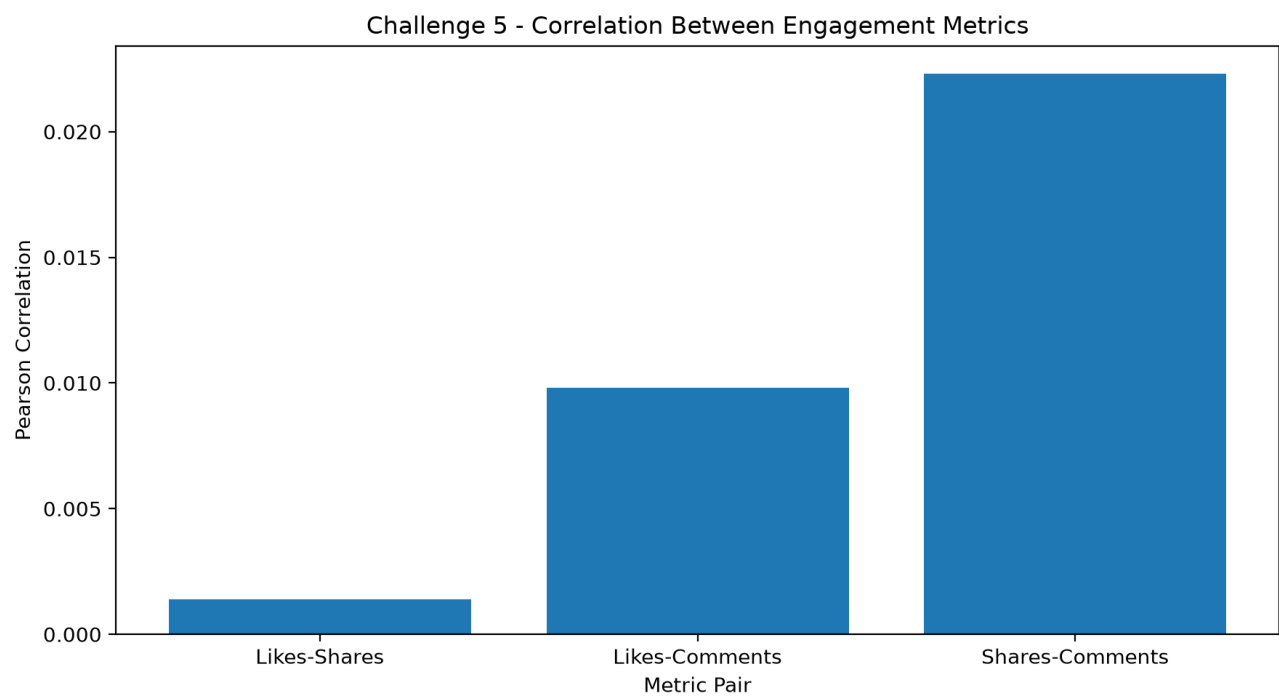
post_id	user_id	platform	total_engagement	rank_within_platform
gmoeib832zbs	user_pe5ycky	facebook	7764	1
a1kiw1618lzy	user_ealari8o	facebook	7683	2
k5qkhdj5zpnw	user_v8sub55d	facebook	7477	3
5dsm9m39dfj3	user_6ftbnlhg	facebook	7455	4
9icy2ek2v4p1	user_ujlji1n7	Challenge 3 - Top 10 High-Engagement Posts per Platform	7419	5
wvm6w4juhyzc	user_c6x0yszg	facebook	7418	6
qrd8bfkdi6wb	user_3318o2cp	facebook	7394	7
u3xsdbdgg81r	user_h6g9kidx	facebook	7280	8
sha7t764lt9t	user_36v8f0yz	facebook	7253	9
br3gu73iot8j	user_bkblavjw	facebook	7159	10
ycj5zzt7mvx	user_d9971ba6	instagram	7893	1
pvf3d8hj7jd	user_cslubwvk	instagram	7738	2
tne7s3o4i4wd	user_lr3fegdi	instagram	7712	3
ukluajoekps5	user_jck2yka1	instagram	7432	4
d4hmvast2kdq	user_5r58l1kr	instagram	7415	5
n8i64t1nwb	user_5805b7s5	instagram	7370	6
q78oevjw0y6	user_jxfroxq0	instagram	7352	7
7f0dwum4w5mi	user_ha9bzru4	instagram	7335	8
mpj2muJ5edfa	user_lpa67a1	instagram	7334	9
zwjyre0ehxsp	user_s87enJ6b	instagram	7312	10
wo7py9alig3t	user_o8ie7hqf	reddit	7793	1
g7d0gdwipoy	user_yhe9m0z0	reddit	7583	2
d7a3bjdzz5w	user_24wzfb8b	reddit	7500	3
xzt681s13lbn	user_w4p8zi5g	reddit	7479	4
3fnsgd3uzxp	user_kk1qdso3	reddit	7435	5
f8agecedqc5x	user_qpmg4srk	reddit	7430	6
atk7729zdm3g	user_rqzqyy9g	reddit	7406	7
0k5m7bn3n3zl	user_lvpej0jp	reddit	7404	8
w3cae3ulmaf	user_uw0vd87k	reddit	7397	9
9ek0fta6yakp	user_hq1ekhdd	reddit	7391	10
sohx1tia2np4	user_716l1rz5	twitter	7440	1
076lqd30r7iu	user_ral75hq0	twitter	7384	2
yhxvwx8u4gqk	user_h8g2fyv0	twitter	7356	3
6d1wkacoavg2	user_t38ni55r	twitter	7317	4
86a89w8sxemc	user_j2p1s57x	twitter	7274	5
5w726ctg6axb	user_bxxatzfk	twitter	7262	6
d9ujzt96mtx8	user_ysobum87	twitter	7251	7
3pdga6na2ohy	user_ztrgkwrt	twitter	7251	8
7j7fd1rh49o3	user_n4b1bcqh	twitter	7226	9
36sulv2r8i4	user_bgppnqks	twitter	7221	10
5kvuyrf38nqx	user_o0feut2e	youtube	7755	1
5n161ir5hhhr	user_u98jwp3f	youtube	7610	2
rh7j9kj7tw8	user_01mu2x1s	youtube	7581	3
ze6nw486cyn2	user_rj884hbd	youtube	7540	4
2ubp2yirtncq	user_q2y6x6ct	youtube	7472	5
s7wyr8sbp28r	user_pp3dgfek	youtube	7444	6
8easa8kz8ahm	user_hq06wwgk	youtube	7392	7
waoid5d12a3s	user_5n6jz0q7	youtube	7383	8
rcxi91rmbvkp	user_j0mgj0v7	youtube	7304	9
pgut5k5d8mji	user_lpjugo20	youtube	7262	10

Challenge 4 — Behavioural User Grouping

Challenge 4 - Behavioural User Grouping

user_group	user_count	avg_posts	avg_engagement
Active User	890	6.62	3994.37
Casual User	496	3.17	3978.1
Power User	114	10.96	4020.77

Challenge 5 — Correlation Analysis



4. Logic Explanation

Challenge 1 — Trend Detection

The query groups posts by year-month using STRFTIME. COUNT(*) measures monthly post volume, while AVG(total_engagement) measures average engagement.

Challenge 2 — Platform Engagement

The query groups posts by platform and calculates post count, average engagement and median engagement for comparison.

Challenge 3 — High-Engagement Detection

ROW_NUMBER() ranks posts separately within each platform by total engagement. Only the top 10 ranks are retained, so the result is dynamically generated.

Challenge 4 — Behavioural User Grouping

A CTE calculates each user's post count and average engagement. CASE then classifies users as Power, Active or Casual users according to posting frequency.

Challenge 5 — Correlation Analysis

DuckDB CORR() calculates Pearson correlation between likes, shares and comments. Values close to zero indicate little linear relationship.

5. Phase 2 Insight Report

- **Monthly activity:** May 2024 had the highest post volume with 761 posts, while February 2025 had the lowest with 667 posts.
- **Engagement trend:** Average engagement fluctuated rather than showing a consistent upward or downward trend. March 2025 recorded the highest monthly average engagement at 4,074.86.
- **Platform performance:** Instagram had the highest average engagement at 4,038.52, while Facebook had the highest median engagement at 4,050.
- **High-engagement posts:** The window-function analysis identified the top 10 posts within every platform. The highest individual engagement value was 7,893 on Instagram.
- **User behaviour:** Active Users formed the largest group with 890 users. Power Users numbered 114 and had the highest average engagement among the groups.
- **Correlation:** Likes-shares correlation was 0.0014, likes-comments was 0.0098, and shares-comments was 0.0223. All are very close to zero.
- **Anomaly screening:** The Phase 1 IQR audit found no total-engagement outliers. Therefore, the high-engagement analysis focuses on relative extremes within each platform rather than claiming they are statistical outliers.

6. Methodology & Reproducibility

All analytical challenges were executed using SQL against the DuckDB database. The outputs were generated directly from the dataset rather than manually entered.